Project Report

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

Team Members

Team ID : LTVIP2026TMIDS61896

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Team Member :3	Bandireddy Dusyanta Venkata Sai Gopinadh

1. INTRODUCTION

1 Project Overview

This project focuses on analyzing housing market trends by examining key attributes such as sale price, age of house, renovation history, number of bedrooms, bathrooms, and floors. The objective is to transform raw housing data into interactive Tableau dashboards that provide meaningful business insights.

The system integrates cleaned housing datasets into Tableau, where visual analytics techniques are applied to explore patterns in pricing distribution, renovation impact, and structural features. By converting static datasets into dynamic dashboards, the project enables real estate stakeholders to make informed decisions regarding pricing strategy and market trends.

Additionally, the dashboard supports web integration using Flask, making it accessible through an interactive website interface.

1.2 Purpose

The main purpose of this project is to:

- Analyze housing price trends
- Identify factors influencing property value
- Study renovation impact on sales
- Visualize patterns using Tableau dashboards
- Support real estate stakeholders with data-driven insights

2. IDEATION PHASE

2.1 Problem Statement

Customer Problem Statement :

I am:
A real estate analyst, investor, or housing market researcher.

I'm trying to:
Understand housing price trends and identify the key features affecting property value.

But:
The data is available in raw spreadsheet format and lacks visual clarity.

Because:
There is no centralized interactive dashboard to analyze relationships between house features and sale price.

Which makes me:
Confused and unable to make confident pricing or investment decisions.

Which makes me...

Overwhelmed, confused, and unable to make informed decisions or draw clear conclusions about global heritage distribution and conservation priorities..

Problem Statement (PS)

- PS-1
  I am a real estate investor
  I'm trying to identify high-value properties and pricing trends
  But the dataset is complex and unorganized
  Because there is no centralized interactive visualization
  Which makes me feel uncertain in decision-making

- PS-2
  I am a housing market researcher
  I'm trying to analyze renovation impact on sale price
  But the data lacks clear visual comparison tools
  Because static reports do not allow dynamic filtering
  Which makes me feel overwhelmed and inefficient

2.2 Empathy Map Canvas

WHO are we empathizing with?

Primary Users:

- Real estate analysts
- Property investors
- Market researchers
- Data analysts
- Housing developers

What do they NEED TO DO?

- Analyze price distribution
- Compare renovated vs non-renovated houses
- Identify relationship between house age and price
- Study impact of bedrooms and bathrooms on price

What do they SEE?

- Large housing datasets in CSV format
- Unstructured numerical data
- No direct visual relationship between features

What do they SAY?

- “We need clear visualization of price trends.”
- “Raw data is difficult to interpret.”
- “We need interactive filters for better comparison.”

PAINS:

- Complex dataset
- No structured visualization
- Manual analysis takes time

GAINS:

- Interactive dashboard
- Quick price comparison
- Data-driven pricing strategy

Solution Statement

The Housing Market Analysis project transforms raw housing data into interactive Tableau dashboards that provide meaningful insights into sale prices, renovation impact, and feature-based value distribution, enabling stakeholders to make data-driven real estate decisions.

2.3 Brainstorming

Brainstorm & Idea Prioritization:

The team conducted brainstorming sessions to identify key analytical aspects of housing data.

Team Leader : Sai Dinesh Kumar Vema

Team member : John Victor Dabbakuti

Team member : Ravipati Lokesh Sai Kumar

Team Member :Bandireddy Dusyanta venakata Sai Gopinadh

Step 2: Idea Listing

Ideas generated:

- Sale price analysis
- Renovation impact study
- Age vs price comparison
- Bedroom/Bathroom influence
- Feature distribution analysis

Step-3: Idea Prioritization

After grouping and prioritizing ideas, the team selected the most impactful features for dashboard development:

- Sale Price Histogram
- Renovation vs Price Comparison
- Age of House Analysis
- Feature-based Distribution

3. REQUIREMENT ANALYSIS

3.1 Solution Requirements

Functional requirements :

S.NO	Requirement ID	Functional Requirement Description
1	FR-01	The system shall import housing dataset.
2	FR-02	The dashboard shall display sale price distribution.
3	FR-03	The system shall analyze renovation impact.
4	FR-04	The dashboard shall provide feature comparison charts.
5	FR-05	The system shall allow dynamic filtering of data.

NON-FUNCTIONAL Requirements :

S.NO	Requirement ID	NON-FUNCTIONAL REQIUREMENT DESCRIPTION
1	**NFR-01**	The dashboard shall load within acceptable time limits.
2	**NFR-02**	The system shall ensure data accuracy.
3	**NFR-03**	The interface shall be user-friendly**.**
4	**NFR-04**	The solution shall be scalable for future dataset updates.

3.2 Flow chart :

The project flow includes:

1. Data Collection
2. Data Cleaning and Transformation
3. Loading Data into Tableau
4. Creating Visualizations
5. Building Dashboard
6. Creating Story
7. Web Integration
8. Deployment

1.Data Collection & Extraction

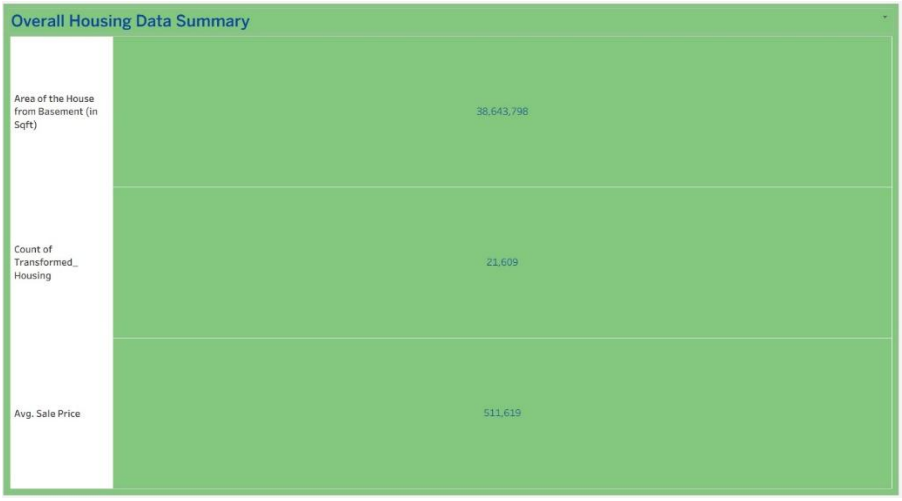
- Collected housing dataset from CSV/Database source.
- Connected dataset to Tableau for analysis.
- Reviewed and understood important columns like sale price, house age, renovation status.

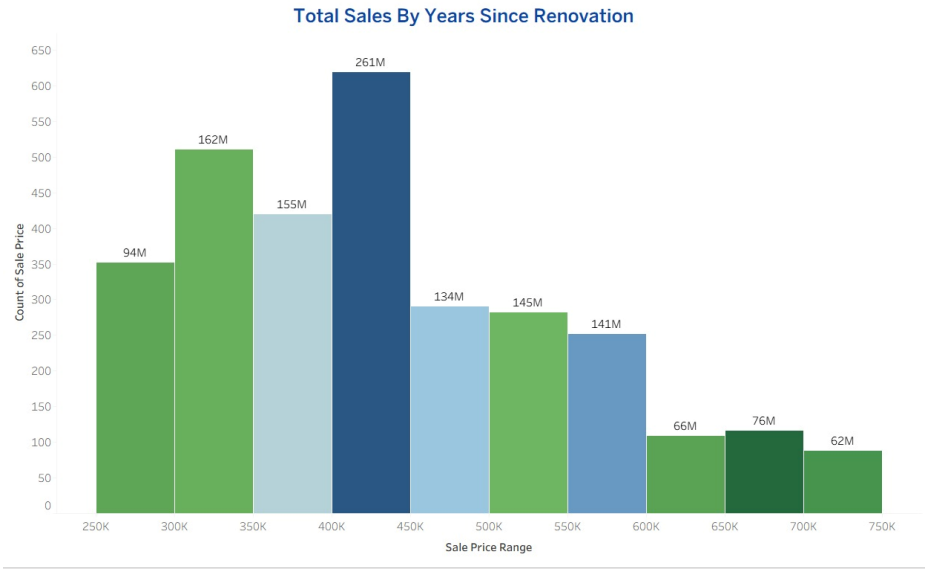
2. Data Preparation

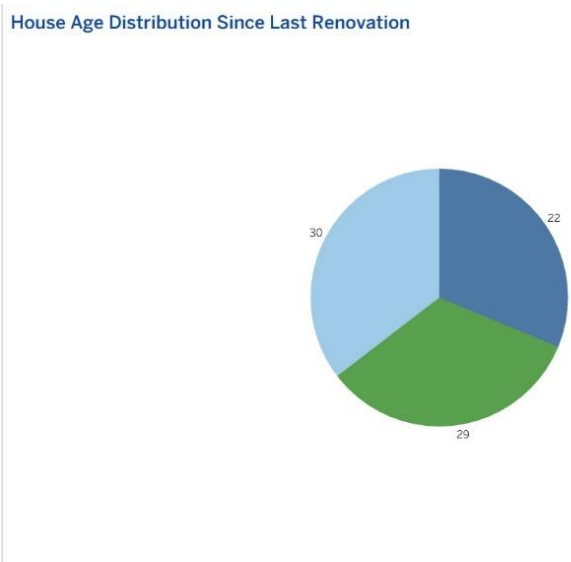
- Cleaned missing or inconsistent data.
- Created calculated fields (e.g., Age of House in Years).
- Structured data properly for visualization.

3. Data Visualization

- Created multiple charts (bar charts, line charts, histograms).
- Analyzed relationships like sale price vs renovation and house age.
- Applied interactive filters (Top N Sale Price, Age filter).

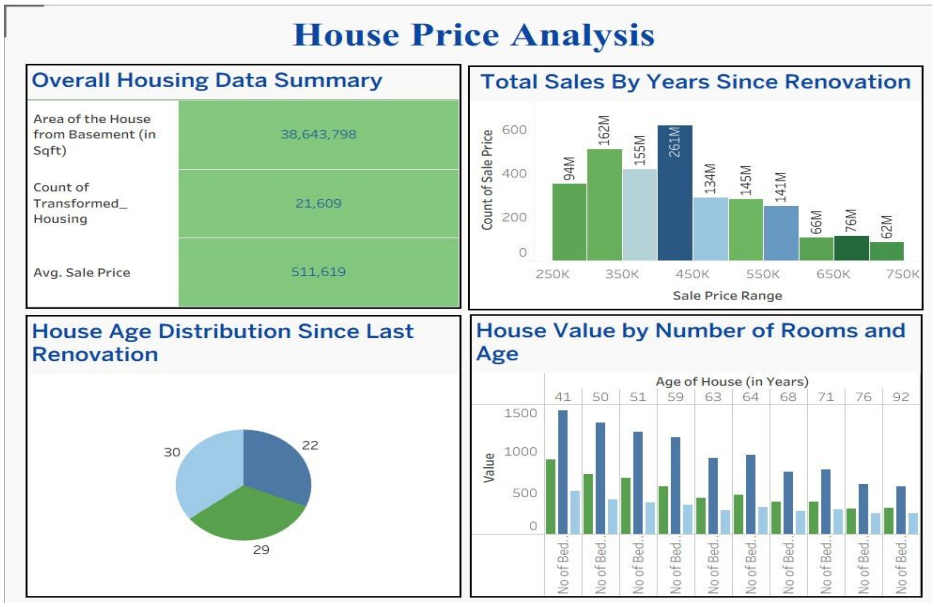






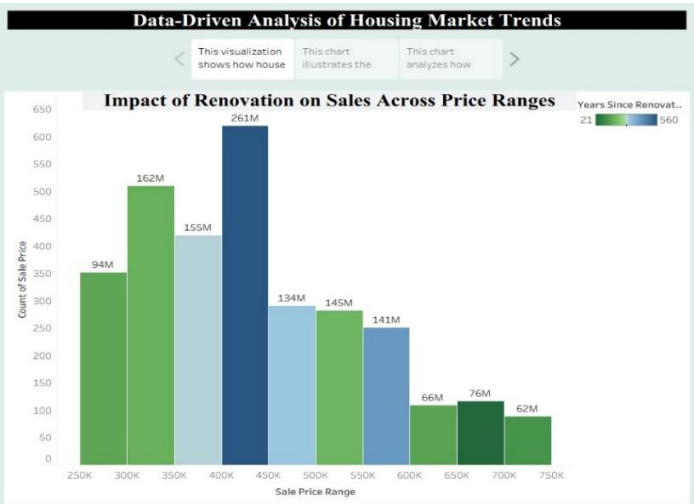
4. Dashboard

- Combined all visualizations into a single interactive dashboard.
- Added filters, KPIs, and dynamic controls.
- Designed responsive layout for better user experience.



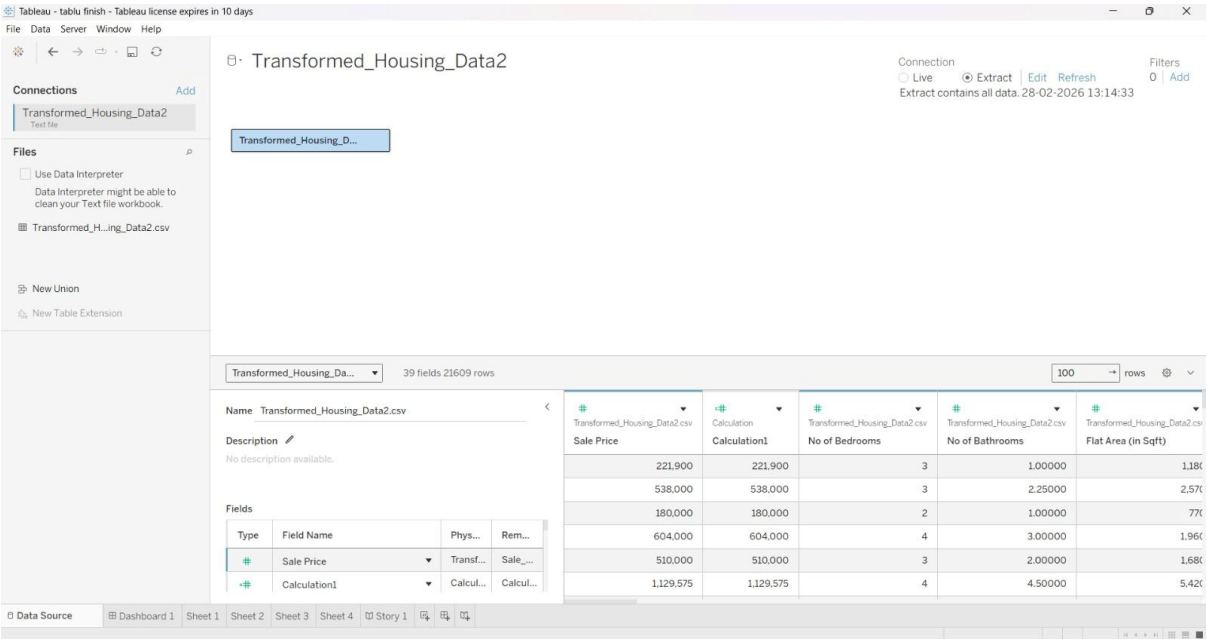
5. Story

- Created sequential story scenes in Tableau.
- Explained housing trends step-by-step with insights.
- Added text descriptions and annotations for clarity.



6. Performance Testing

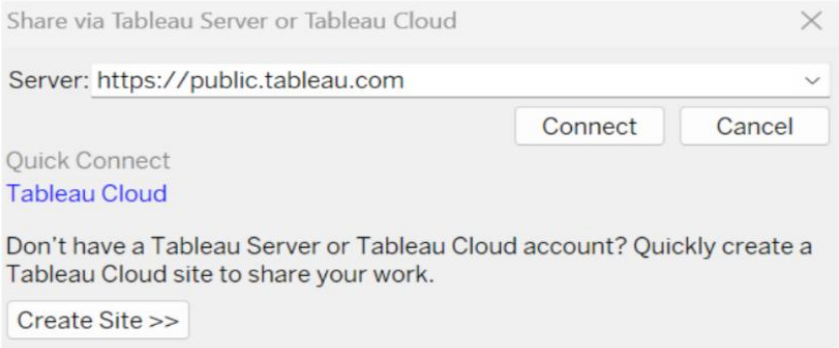
- Tested dashboard performance with full dataset.
- Optimized calculated fields and filters.
- Ensured smooth loading and fast interaction.



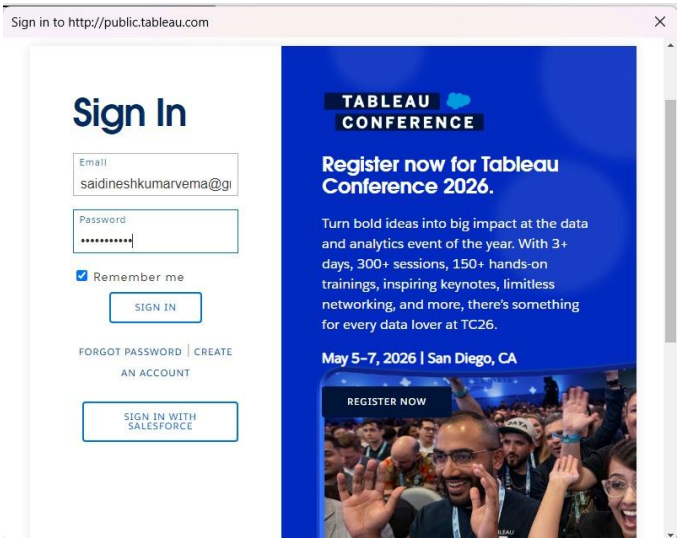
7. Web Integration

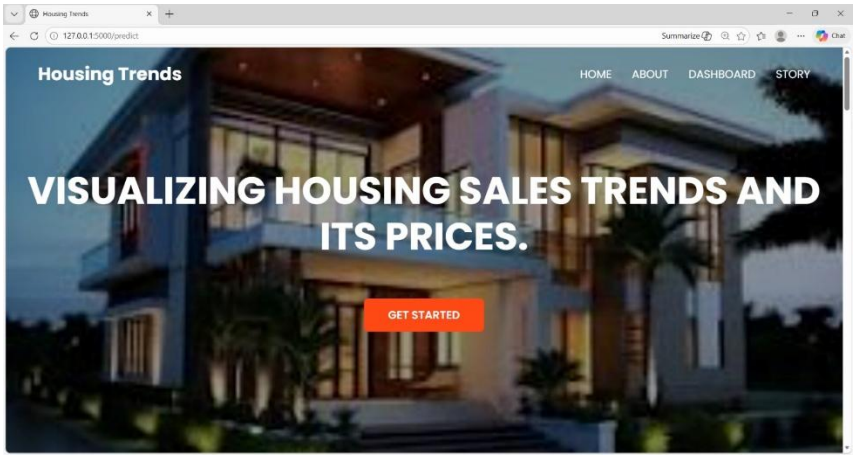
- Published dashboard on Tableau Public.
- Embedded dashboard into web interface using Flask.
- Enabled online access and sharing of insights.
- **Publishing**

  Give the server address of your tableau public account and click on connect.



Step 2: Once you click on connect it will ask you for tableau public user name and password.





8. Project Demonstration & Documentation

- Recorded end-to-end explanation video.
- Prepared detailed project documentation.
- Included architecture, workflow, and screenshots.

3.3 Technology Stack

- Data Source: CSV Housing Dataset
- Data Processing: Excel
- Visualization Tool: Tableau
- Web Framework: Flask
- Deployment Platform: Render

4.Proposed Solution

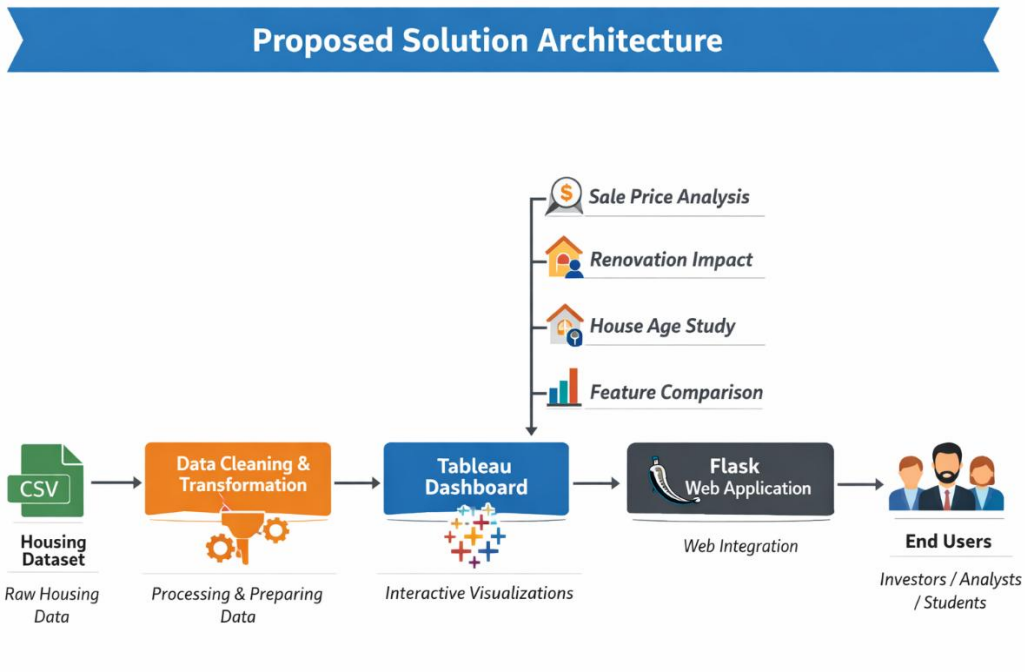
4.1 Problem–Solution Fit

The proposed solution is to develop an interactive Tableau dashboard that visualizes housing market data.

The dashboard enables users to:

- Analyze sale price distribution
- Compare renovation impact
- Study house age influence
- Evaluate room-based price variation

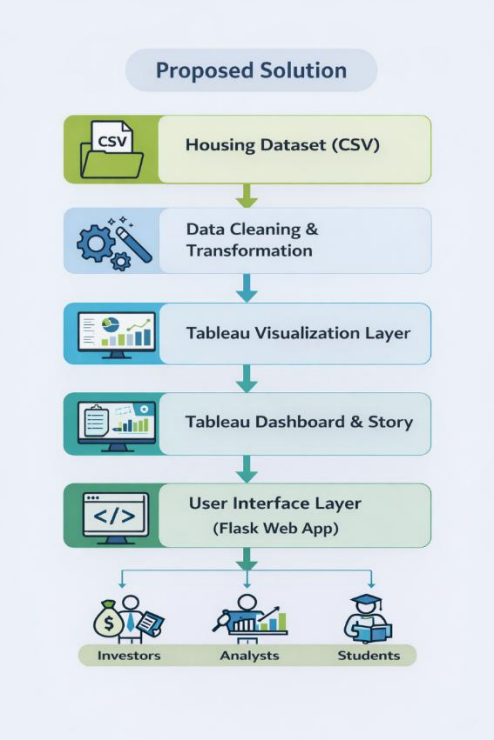
The solution is unique because it transforms raw housing data into structured, interactive visual analytics instead of static reports.



5. SOLUTION ARCHITECTURE

5.1 Steps in Solution

- The architecture begins with a structured housing dataset containing property details.
- The data undergoes cleaning and preprocessing.
- The processed dataset is loaded into Tableau.
- Interactive dashboards and stories are created.
- The dashboards are published to Tableau Public.
- Finally, the dashboard is embedded into a Flask web application for public access.



6. PROJECT PLANNING & SCHEDULING

6.1 The project was divided into two main sprints:

Sprint 1

- **Data Collection**

  Data collection is the process of gathering and measuring information on variables of interest, in an established systematic fashion that enables one to answer stated research questions, test hypotheses, evaluate outcomes and generate insights from the data.

- **Data Cleaning**

  Preparing the data for visualization involves cleaning the data to remove irrelevant or missing data, transforming the data into a format that can be easily visualized, exploring the data to identify patterns and trends, filtering the data to focus on specific subsets of data, preparing the data for

  visualization software, and ensuring the data is accurate and complete

Sprint 2

- ### Dashboard Creation

  Data visualization is the process of creating graphical representations of data to help people understand and explore the information. The goal of data visualization is to make complex data sets more accessible, intuitive, and easier to interpret. By using visual elements such as charts, graphs, and maps, data visualizations can help people quickly identify patterns, trends, and outliers in the data.

- ### Story Development

  A storyboard is a visual representation of a sequence of events, typically used in multimedia projects such as films, animations, advertisements, or presentations. It consists of a series of drawings or images arranged in a sequence, often accompanied by annotations or descriptions, to outline the flow of the story or concept

- ### Web Integration

  Publishing helps us to track and monitor key performance metrics, to communicate results and progress. help a publisher stay informed, make better decisions, and communicate their performance to others. Publishing dashboard and reports to tableau public

- ### Deployment

  The Tableau dashboard and story were successfully integrated into a web application using embedding techniques.The application was deployed to ensure users can access interactive visualizations seamlessly through a web browser.

Sprint Table:

Sprint	Taks Included	Description
Sprint 1	Data Collection	Gathered housing dataset from reliable sources
	Data Cleaning	Handled missing values, removed duplicates, corrected inconsistencies
	Feature Engineering	Created new features such as house age, renovation indicators, etc.
	Data Transformation	Structured dataset suitable for visualization and analysis
Sprint 2	Dashboard Creation	Developed interactive Tableau dashboard with multiple charts
	Story Development	Created storytelling sheets to explain insights clearly
	Web Integration	Embedded Tableau dashboard into web application
	Deployment	Hosted the application for user access

7. PERFORMANCE TESTING

7.1 Performance metrics analyzed:

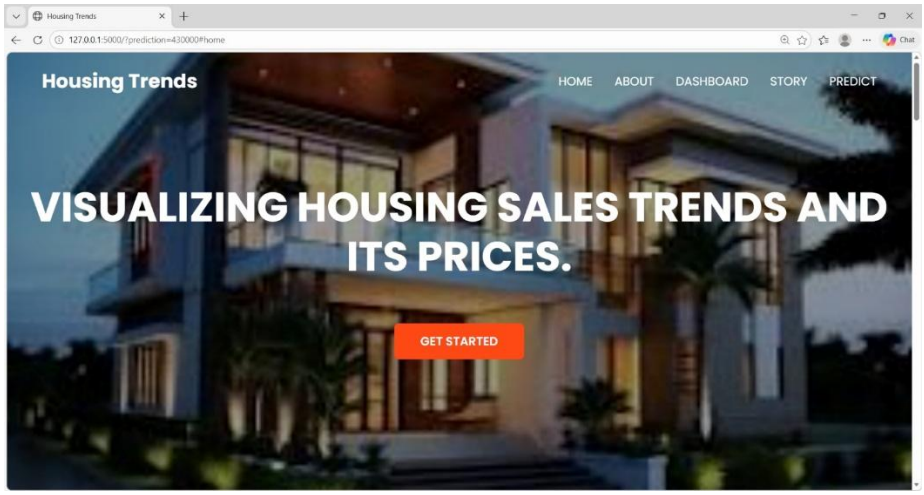
- Dashboard Load Time
- Filter Response Time
- Visualization Rendering Speed
- Mobile Responsiveness

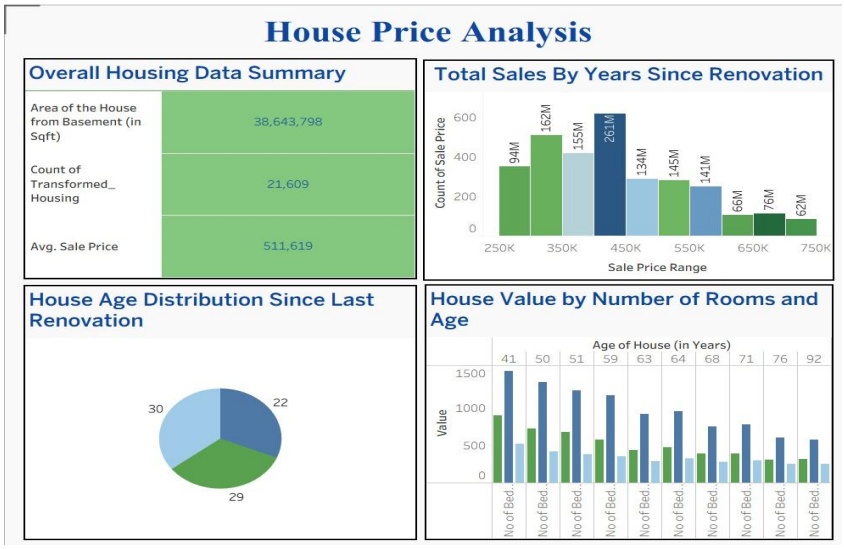
  The dashboard performed efficiently with acceptable load time and smooth interaction on desktop devices.

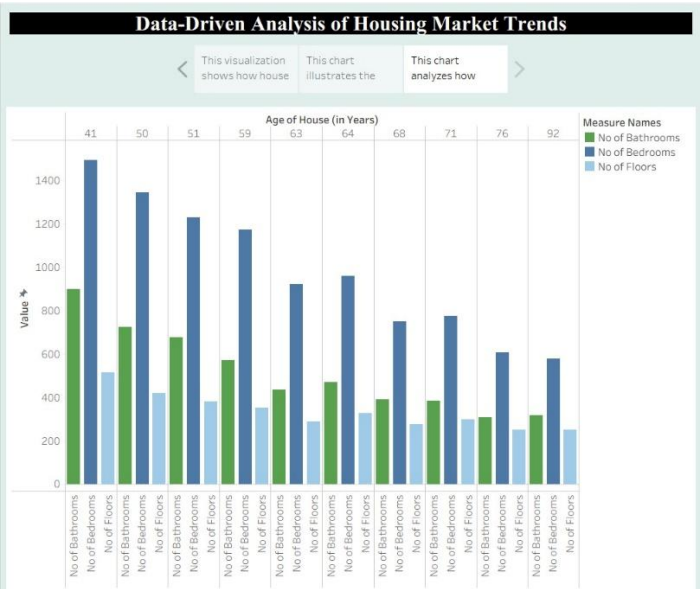
8. RESULTS

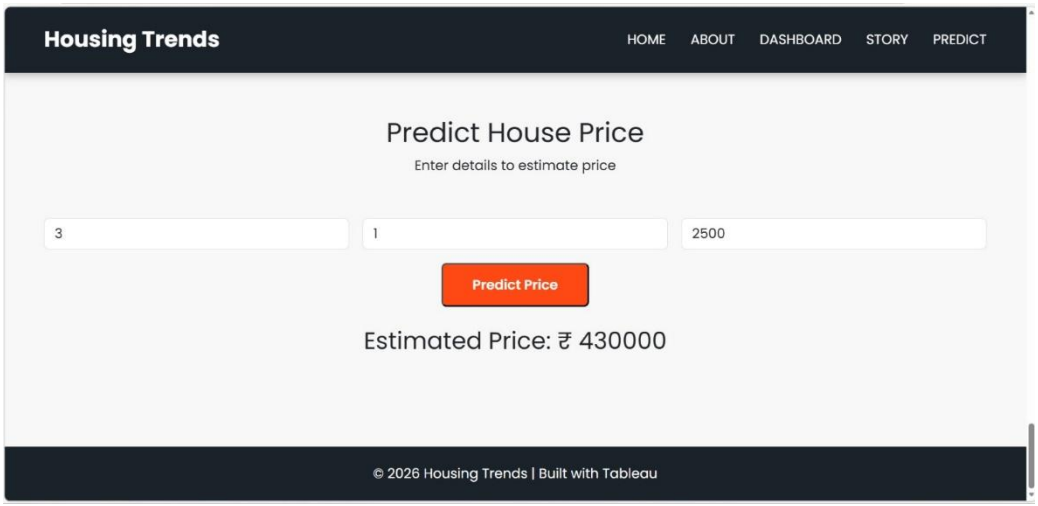
8.1 Key findings from the analysis:

- Mid-range properties dominate sales distribution.
- Recently renovated houses show higher pricing trends.
- Number of bedrooms significantly affects sale price.
- House age impacts property value variation.


Housing Trends
127.0.0.1:5000/?prediction=430000#home
Housing Trends
HOME
ABOUT
DASHBOARD
STORY
PREDICT
VISUALIZING HOUSING SALES TRENDS AND ITS PRICES.
GET STARTED


House Price Analysis
Overall Housing Data Summary
Area of the House from Basement (in Sqft)
38,643,798
Count of Transformed_ Housing
21,609
Avg. Sale Price
511,619
Total Sales By Years Since Renovation
Count of Sale Price
600
400
200
0
94M
162M
155M
261M
134M
145M
141M
66M
76M
62M
250K
350K
450K
550K
650K
750K
Sale Price Range
House Age Distribution Since Last Renovation
30
22
29
House Value by Number of Rooms and Age
Age of House (in Years)
41
50
51
59
63
64
68
71
76
92
Value
1500
1000
500
0
No of Bed.


Data-Driven Analysis of Housing Market Trends
This visualization shows how house
This chart illustrates the
This chart analyzes how
Age of House (in Years)
41
50
51
59
63
64
68
71
76
92
Measure Names
No of Bathrooms
No of Bedrooms
No of Floors
Value
1400
1200
1000
800
600
400
200
0



9. ADVANTAGES

9.1 Outcomes in this project

- Interactive visualization
- Faster insight discovery
- Easy comparison of property features
- User-friendly interface
- Professional dashboard design

10. CONCLUSION

The Housing Market Analysis project successfully demonstrates the power of data visualization using Tableau. The project converts complex housing data into interactive dashboards that support real estate decision-making.

The integration with Flask enhances accessibility by allowing online interaction with the dashboard.

Overall, the project provides a scalable, efficient, and insightful framework for housing market analysis.

11. FUTURE SCOPE

Future enhancements may include:

- AI-based price prediction
- Real-time data integration
- Advanced forecasting models
- Mobile-optimized dashboards
- Integration with real estate APIs

All Links

Data set : **House Price Prediction Dataset**

Tableau Viz Public URL : **Dashboard**

Project Demo Live link : **Demo link**